Network Design for Faculty of Computing Block N28c

TASK 6B: MAKING AN INDIVIDUAL REPORT

SECR1213 - NETWORK COMMUNICATIONS

SEMESTER I, SESSION 2024/2025

SECTION 02

Lecturer: Dr. ISMAIL FAUZI BIN ISNIN

Group Name: Network Nexus

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Reflection Video: https://drive.google.com/drive/folders/1JaE8-8l1qn54baf51QLtQIjtJi818Y67?usp=drive_link

• Detail your contribution to the group and the project work

• Detail the other members contribution to the group and the project work

My group has very clear task distribution for each task. We will have a meeting before each task.

For task 1, since Naim still not join our group, the task distribution is around me, Zhi Ming and Yu Hin. I suggested all facilities that should be existed in the building and then assigned the designing task around us. I took responsibility to design the General Labs, Cisco Networking Lab, Hybrid Classroom and Store Room, Zhi Ming took the Student Lounge, Video Conferencing Room, Server Room and Database Room, while Yu Hin took the Embedded System Lab, Network Operation Center, Meeting Room and Maintenance Room. For task 2, Naim already joined our group. We both suggested 10 questions for preliminary analysis, then we gathered together to select the most suitable question. Finally, we had selected 13 questions. For feasibility analysis part, I was responsible for the technical feasibility, Zhi Ming for operational feasibility and Yu Hin responsible for finance feasibility. For task 3, we both had a meeting together to discuss the devices needed in the building. We compared the price and functionality between various devices and select the most suitable device focusing on their performance and sustainability. I was responsible for the devices in General Lab, Cisco Networking Room and Server Room. For task 4, I task responsibility in designing the network cabling for Meeting Room, and Maintenance Room, and also the overall network distribution diagram. While for my teammates, Zhi Ming is responsible for the two General Labs and Video Conferencing Room, Yu Hin is responsible for the Hybrid Classroom, Server Room and Database Room, while Naim is responsible got the Network Operation Center, Embedded System Lab and Cisco Networking Lab. For task 5, Naim and Zhi Ming did the calculation of subnetting, Yu Hin and I assign the IP address for device for each facility. For task 6, I create a need-to-do list, I was responsible for the introduction part and also write our own responsibilities.

• Explain how you work as a group.

Before each task, we will have a short face-to-face meeting for task distribution and discussion. During each task, we will have extra meeting if one of my teammates got any problem. For myself, I studied the task detail before joining each meeting to make sure what I need to do. Then, I did pre-research for every task to collect idea so that I can present it to my teammate during each meeting. Moreover, I keep updating my task progress in my team and finishe• Explain what you have learned from working as a group.

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I learn how important of communication and problem-solving skill in a project. Since we only got limited time for each task, we need to have a very clear task distribution and also need to know in very clear what should we do. So we actively conducting meeting for each task discussion. This process enhances my communication and problem-solving skill effectively.

• Explain what you have learned from doing the project.

In technical aspect, the first thing I learn is what basic network devices should be install in a building to ensure stable networking connection. I keep doing research on the function of networking devices. Some key devices include router – a device to send packets from one to others until reach the destination, switch – a device that centralizes communications among several connected devices, and server – a device to serve data for clients in a client-server model. Moreover, I learn how to figure the network address, subnet mask and also assigning of IP address.

• Your comments and suggestions on the project.

I think this project is quite interesting because we need to design the network infrastructure for FC's new building. I have a sense of expectation for this project because I'm also a student in FC, so I will think is it possible my team solution taken by FC? So, is it possible I can step into this building in the future? I'm a little bit looking forward for it.

Furthermore, this project is great because it let me to learn many network devices, including router, switch, access point, modem, firework, server and so on. I also learn how these devices can connect to each other perform a complete network architecture.

As my suggestions, I hope I can have hands on activity on the device connection, such as configuring routers and switches, as well as setting up the network physically and virtually. This practical activity will enhance my understanding of how theoretical knowledge can be applied in real-world scenarios.